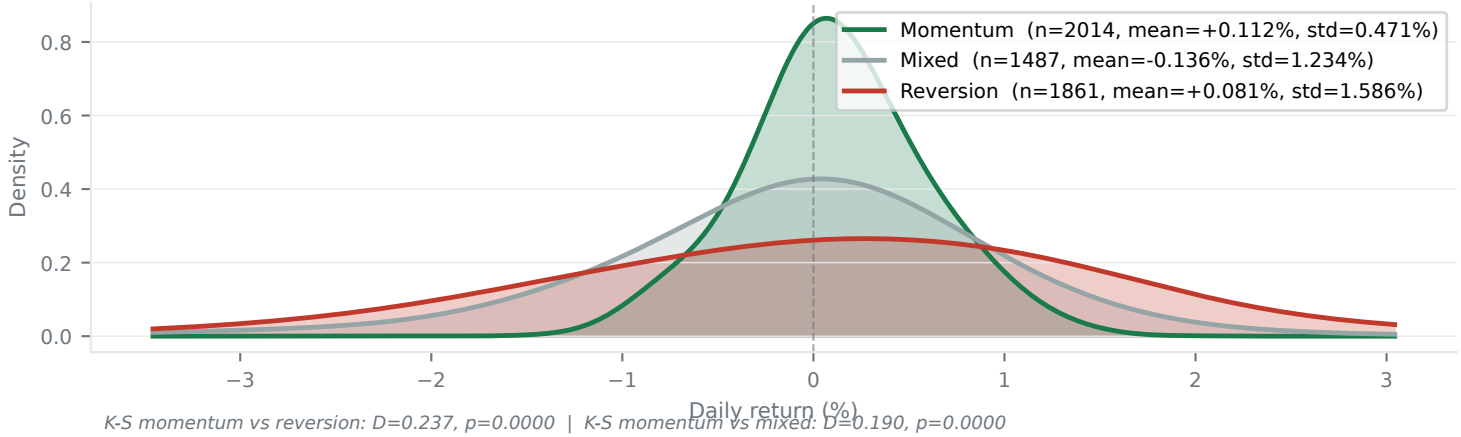
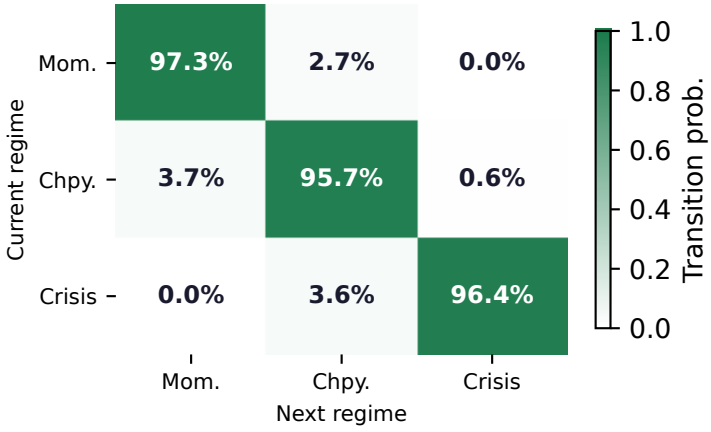
Regime Ensemble - Technical Report

Return Distributions by Regime (Kernel Density Estimate)



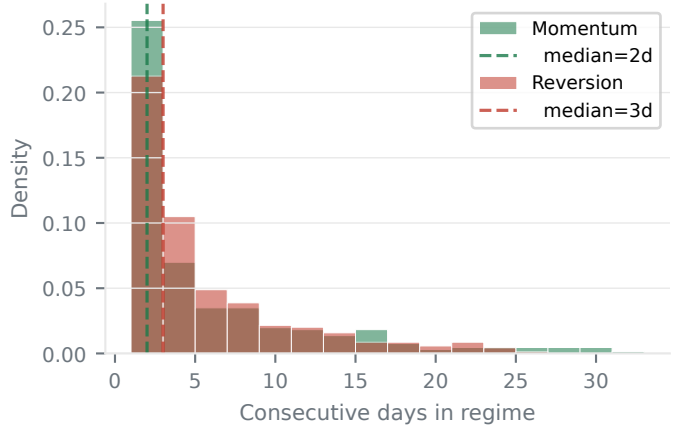
K-S momentum vs reversion: D=0.237, p=0.0000 | K-S momentum vs mixed: D=0.190, p=0.0000

Markov Transition Matrix



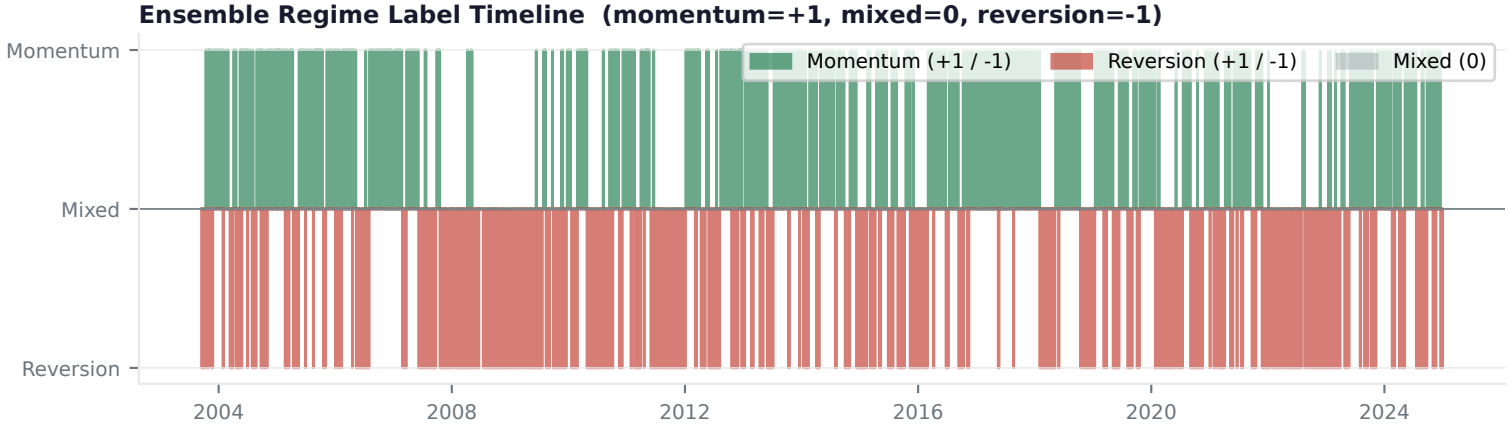
Diagonal = regime persistence. High diagonal = sticky regimes.

Regime Sojourn Time Distribution



Short sojourn -> fast flip -> higher turnover -> cost drag.

Ensemble Regime Label Timeline (momentum=+1, mixed=0, reversion=-1)



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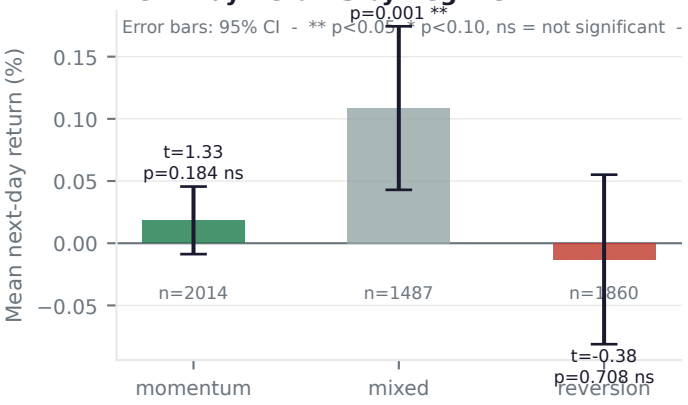
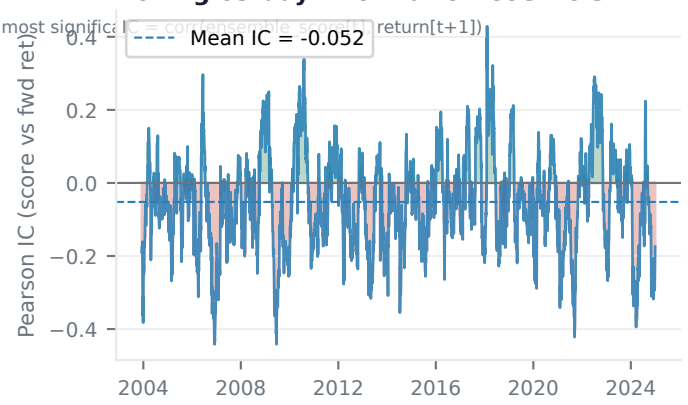
Next-Day Returns by Regime

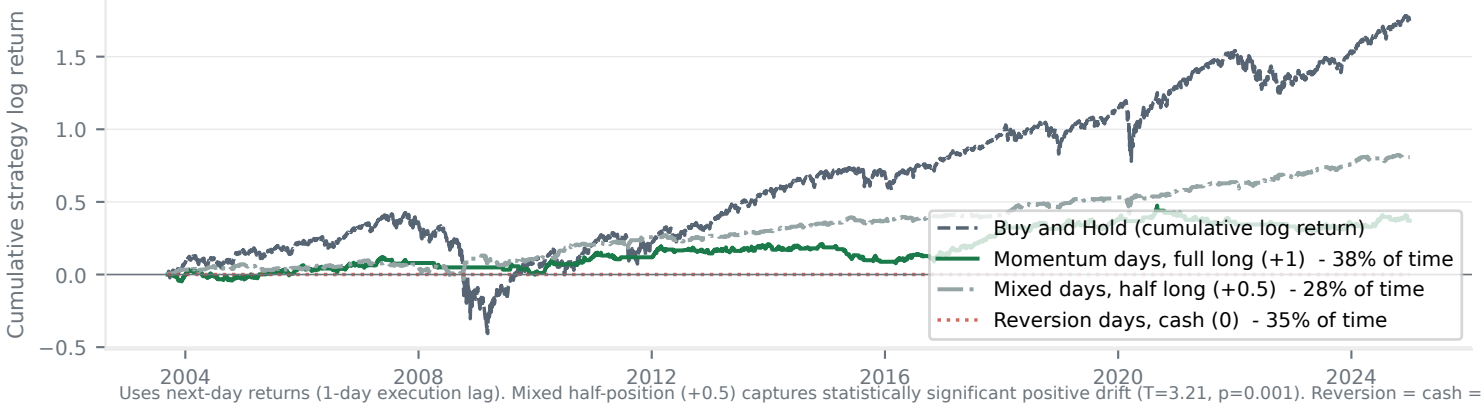
Error bars: 95% CI - ** p<0.05 * p<0.10, ns = not significant - Mixed is the most significant



Rolling 63-day Information Coefficient



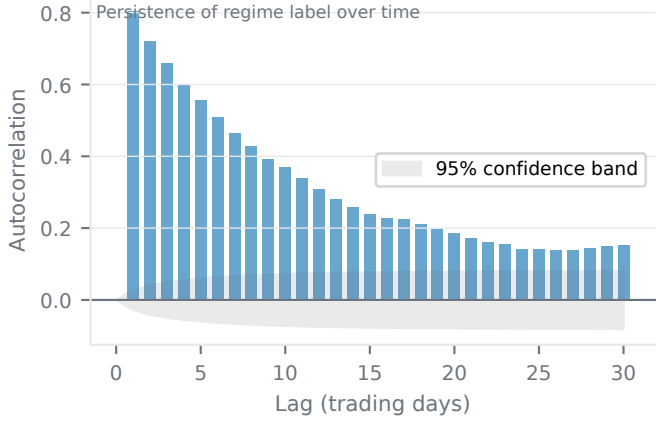
Cumulative Strategy Return Decomposition - Contribution by Regime



Uses next-day returns (1-day execution lag). Mixed half-position (+0.5) captures statistically significant positive drift (T=3.21, p=0.001). Reversion = cash = flat line.

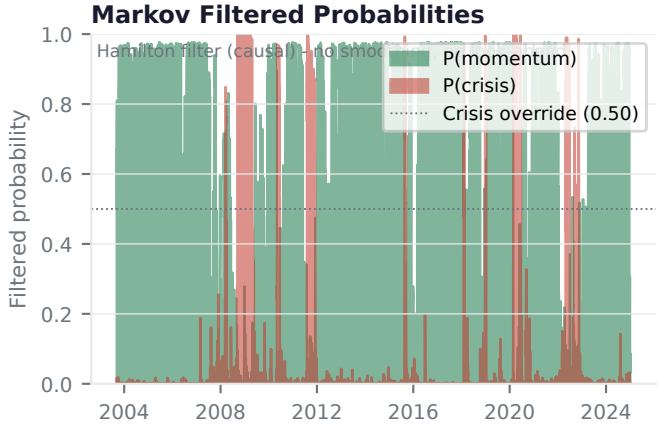
Regime Label Autocorrelation (ACF)

Persistence of regime label over time



Markov Filtered Probabilities

Hamilton filter (causal) - no smoo



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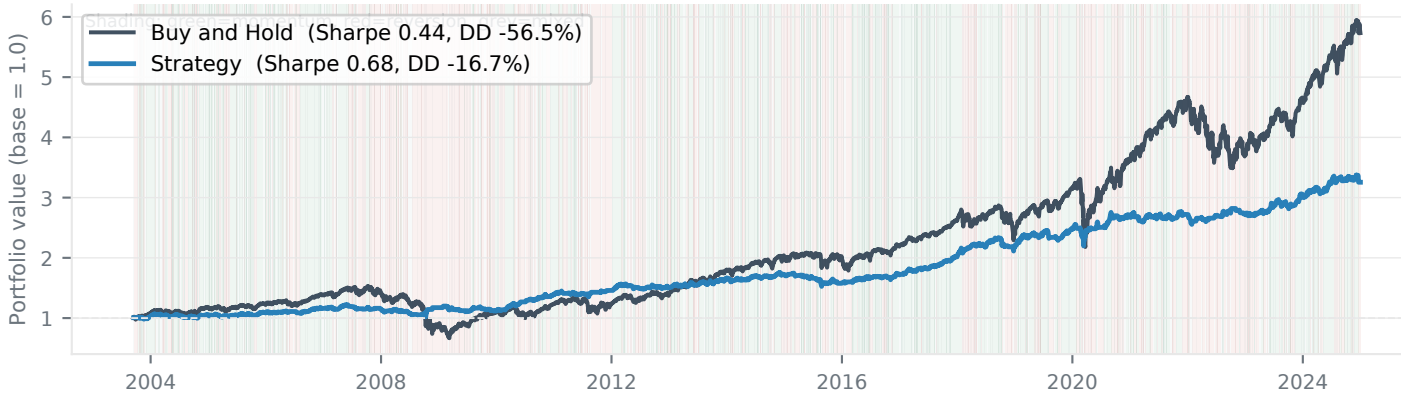
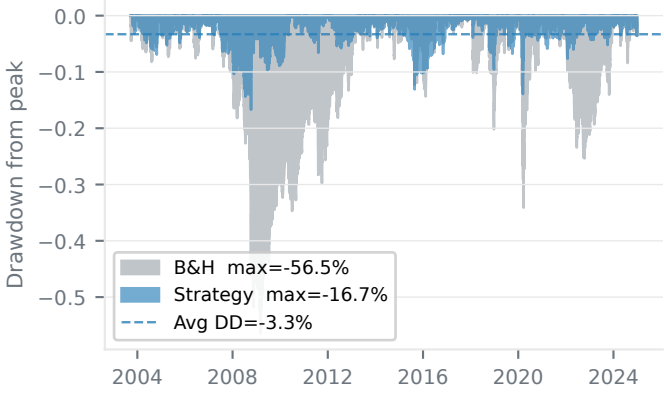
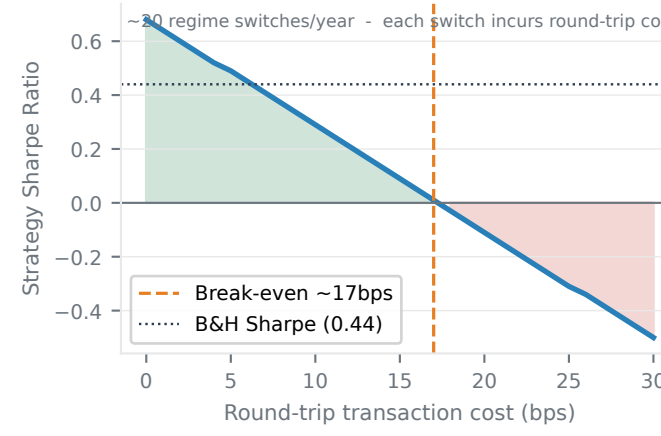
Equity Curves - Regime-Shaded Background (0 bps, 1-day lag)



Underwater Drawdown



Sharpe vs Transaction Cost (0-30 bps)



Full Performance Attribution (SPY 2000-2025, 0 bps)

Metric	Strategy	Buy and Hold
CAGR	**+5.4%**	+6.7%
Vol (ann.)	**8.1%**	18.7%
Sharpe	**0.66**	0.36
Sortino	**0.69**	0.43
Calmar	**0.31**	0.11
Max DD	**-17.1%**	-60.7%
Win rate	**36.2%**	54.7%
Profit fctr	**0.93**	0.90
Information Ratio	**-0.18**	n/a
Mean Exposure	**51.4%**	100.0%
% Time Fully Long	**37.6%**	100.0%
% Time Half Long	**27.7%**	0.0%